

WEINCE

Replace InfoNCE logits with a geometry-adaptive blend

$$\ell_{ij} = (1 - \lambda_i) \underbrace{\frac{s_{ij}}{\tau}}_{\text{Gumbel (InfoNCE)}} + \lambda_i \underbrace{(-\hat{\beta}_i \log(1 - s_{ij}))}_{\text{Weibull (shortfall)}}$$

Loss is standard cross-entropy over ℓ_{ij} . **No new learnable parameters.**

$$\mathcal{L}_{\text{WEINCE}}(\theta) = -\frac{1}{2N} \sum_{i=1}^{2N} \log \frac{\exp(\ell_{i,i+})}{\sum_{j \in \mathcal{N}_i} \exp(\ell_{ij})}$$